

Question	Marking Guidance	Mark	Comments
02.1	1. (Some bacteria have) alleles for resistance; 2. (Exposure to) antibiotics is the selection pressure OR Non-resistant bacteria die OR Resistant bacteria survive/reproduce; 3. More antibiotics used in hospital (compared with elsewhere) OR Patients have weakened immune systems OR (So) high frequency of resistance allele (in bacterial population);	3 (3 x AO1)	1 and 2 Reject reference to immunity only once 1. and 3. Accept gene for allele 1. Reject if antibiotics cause production of resistance gene/allele 2. Accept strain for bacteria 3. Ignore antibiotics prescribed when not needed OR antibiotic course is not finished 3. Ignore defence system, for immune system 3. Accept proportion/percentage for frequency
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02.2	Maltose;	1 (AO1)	Reject maltase Accept phonetic spelling eg moltose/maltosse/maltoze/moltoes/maltoez

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02.3	1. Wash hands with soap OR Disinfect surfaces; 2. Use sterile pipette/syringe (to transfer bacteria); 3. (Remove bottle lid and) flame neck of bottle; 4. Lift lid of (agar) plate at an angle; 5. Work close to upward air movement; 6. Use sterile spreader; 7. Place pipette/spreader into disinfectant (immediately after use);	3 max (3 x AO1)	1. Ignore sterilise hands OR surfaces 1. Accept sanitise for disinfect 1. Accept antiseptic /antimicrobial/alcohol (wipes) 1. and 7 Accept a named type of disinfectant 2. Reject loop 2. Accept use unopened pipette/syringe for sterile 4. Accept lift lid slightly OR keep lid over plate 4. Ignore work quickly with lid off 5 Reject air movements sterilise air 5 Accept convection current for air movement 6. Accept loop for spreader 6. Examples of sterilising technique eg, flame OR dip in alcohol and flame OR dip in disinfectant and rinse (in sterile water)

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02.4	For - Resistant bacteria grow faster with trehalose; (So) resistant bacteria (likely to) increase in frequency in the population/people; - Resistant bacteria (likely to) outcompete non-resistant bacteria; Against - In laboratory not in people; - Other disaccharides (in the diet) might affect bacteria; - Other bacterial species (in the body) might affect bacteria; - No stats test to see if difference/increase is significant; - No data for both resistant and non-resistant bacteria growing together; - No data for different concentrations of trehalose;	3 max (3 x AO3)	Max 2 if only 'For' or only 'Against' marks 1, 2, 3, 4 and 5 Accept C. <i>difficile</i>/strain for bacteria 2. Accept 'percentage/proportion' for 'frequency' 5. Accept carbohydrate OR polysaccharide OR sugar, for disaccharide 7. Accept 'is not due to chance' for 'is significant' 7. Ignore standard deviation/SD (as not a stats test) 7 Reject 'to see if results are significant'

